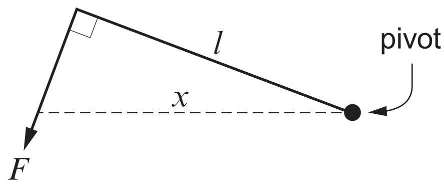
Examiner only

2. (a) The following question has been set in a previous Physics exam:

Explain what is meant by the moment of a force about a pivot.

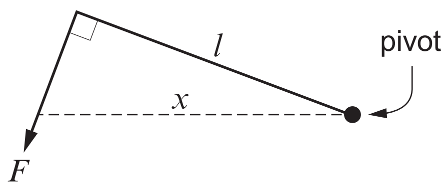
Five of the responses given are shown below (**A** to **E**). **Only three of the responses are correct**. Place a tick (✓) in the boxes next to the correct responses. [2]

A.



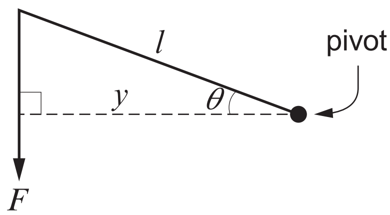
moment = Fl ☐

B.



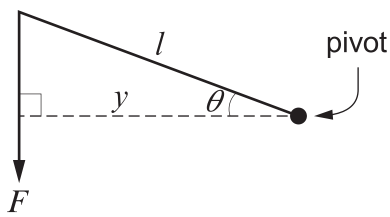
moment = Fx ☐

C.



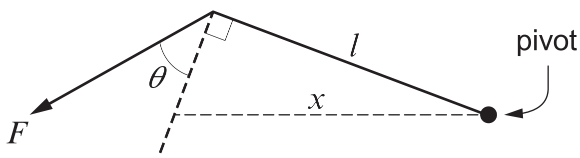
moment = $Fl \cos\theta$ ☐

D.



moment = Fl ☐

E.



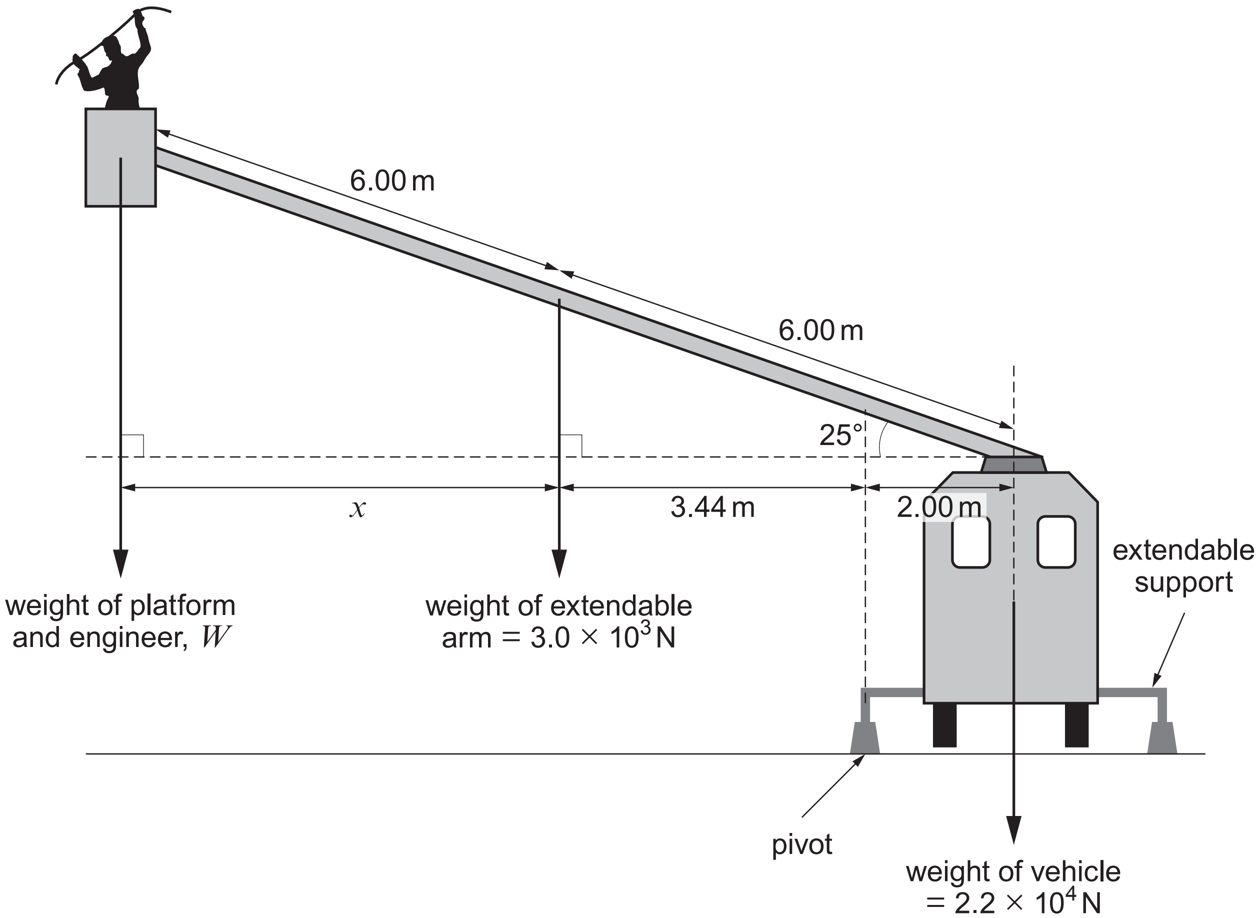
moment = $F \cos\theta\, l$ ☐

2420U101 03

 (2420U10-1) **Turn over.**

Examiner only

(b) Telephone engineers sometimes use an extendable arm to repair telephone lines. The arm is able to extend from the top of a specially modified vehicle. When the arm is in use extendable supports are used as shown. The pivot point is also shown.



(i) Show that the distance, x, in the diagram is approximately 5.4 m. [1]

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(ii) Determine the maximum possible weight of the platform and engineer, W, in the situation shown in the diagram, so that the system does not topple. [3]

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Examiner only

(iii) A sign placed on the platform gives the following warning:

Maximum 2 persons or 200 kg

Consider whether or not this is a reasonable warning to give. Assume the arm is at its maximum extension in the diagram and the angle shown (25°) is the lowest possible. The weight of the platform is 1700 N. [4]

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